

Statistical Mechanics

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1 Boltzmann Distribution

Ex: Spin 1/2 particles take spin values $\sigma \in \mathcal{X} = \{-1, 1\}$. The energy of a particle with spin σ in a magnetic field B is $E(\sigma) = -\sigma B$. The probability density is $p_\beta(\sigma) = \frac{1}{Z(\beta)} e^{-E(\sigma)} = \frac{1}{Z(\beta)} e^{\beta B \sigma}$. We have $Z(\beta) = e^{\beta B} + e^{-\beta B} = 2 \cosh \beta B$. The expected spin, the magnetization, is given by

$$\mathbb{E}[\sigma] = \frac{e^{\beta B} - e^{-\beta B}}{e^{\beta B} + e^{-\beta B}} = \tanh \beta B.$$

Ex: Some spin variables can have a larger space. Potts spin: $\mathcal{X} = [q]$. For magnetic field of intensity h pointing in $r \in [q]$, the energy is $E(\sigma) = -B \delta_{\sigma, r}$. The UPDF is $\tilde{p}_\beta(\sigma) = e^{-\beta E(\sigma)} = e^{\beta B \delta_{\sigma, r}}$. We have $Z(\beta) = \sum_{\sigma \in [q]} \tilde{p}(\sigma) = \sum_{\sigma \in [q]} e^{\beta B \delta_{\sigma, r}} = e^{\beta B} + (q-1)$. Then

$$p_\beta(\sigma) = \begin{cases} \frac{e^{\beta B}}{e^{\beta B} + (q-1)} & \text{if } \sigma = r, \\ \frac{1}{e^{\beta B} + (q-1)} & \text{if } \sigma \neq r. \end{cases}$$

Thus,

$$\begin{aligned} \langle \sigma \rangle &= \frac{e^{\beta B}}{e^{\beta B} + (q-1)} r + \frac{q(q-1)/2 - r}{e^{\beta B} + (q-1)} \\ &= \frac{e^{\beta B} r + q(q-1)/2 - r}{e^{\beta B} + (q-1)} \end{aligned}$$

Note for $T \rightarrow \infty$, ie $\beta \rightarrow 0$, we have $\langle \sigma \rangle = (q-1)/2$, ie the average of the numbers in $[q]$, as expected.

Consider a bottle $\mathcal{X} \leq \mathbb{R}^3$. The energy of $x \in \mathcal{X}$ is $E(x) = wh(x)$, where h is the height of x and w a constant. The ratio of the probabilities of particles at positions x, y is

$$p_\beta(x)/p_\beta(y) = \frac{\exp(-\beta wh(x))}{\exp(-\beta wh(y))} = \exp(\beta w(h(y) - h(x))).$$

so if y is higher than x , then this ratio is greater than 1.

$n \times n$ Chessboard. Let p, q, r denote probabilities of being in a center, edge, and corner, resp. Then $p : q : r = 4 : 3 : 2$. There are $(n-2)^2, 4(n-2), 4$ such squares. Let $p/4 = q/3 = r/2 = t$. Thus, $4(n-2)^2t + 12(n-2)t + 8t = 1$, so $t = 1/(4(n-2)^2 + 12(n-2) + 8) = \frac{1}{4n(n-1)}$. Thus, $(p, q, r) = \left(\frac{4}{4n(n-1)}, \frac{3}{4n(n-1)}, \frac{2}{4n(n-1)}\right)$.

However, we just need the ratios. Corner have highest energy, then edge, then center. We have $f(x) = e^{-\beta E(x)}$. The partition function is $4(n-2)^2 + 12(n-2)t + 8t = 4n(n-1)$.

Claim: $\frac{d^2}{dx^2} f(x) \geq 0$ iff $\frac{d^2}{du^2} f(1/u) \geq 0$.

Suppose $\frac{d^2}{dx^2} f(x) \geq 0$. Then